

Jiya Music App — Full Stack Specification

Comprehensive Technical Architecture, Infrastructure, APIs & Android Native Engine Report

Project Overview: Jiya Music is a high-performance, progressive full-stack music streaming platform built with Next.js 14, TypeScript, Prisma ORM, and SQLite. It integrates a 4-in-1 Universal Music Search & Streaming Engine (JioSaavn 320kbps, YouTube/Invidious, iTunes, Spotify) and features a standalone native Android application compiled in Android Studio with a Java MediaPlayer foreground service for 100% background and screen-off playback.

1. System Deployment & Metadata

Property	Specification / Resource Value
Application Name	Jiya Music (The Music App)
Live Production URL	https://jiya-kappa.vercel.app/
GitHub Code Repository	https://github.com/NirmalyaCh005/JIYA--The-Music-App.git
Web Hosting Platform	Vercel Serverless Edge Platform (Global CDN)
Android App Workspace	Standalone Native Project Directory (/android-app)
Compiled Android Package	Jiya-Music.apk (6.64 MB Standalone APK)

2. Full-Stack Technology Spectrum

Architecture Layer	Technology Stack	Role & Implementation Details
Frontend Framework	Next.js 14.2 (App Router)	Server Side Rendering (SSR), Client Components, React 18 Concurrent Features
Language & Typing	TypeScript 5.x	Strict end-to-end type safety across API routes, data models, and player state
Styling System	Tailwind CSS 3.4	Custom dark-theme design tokens, glassmorphism, responsive UI layouts
UI Components & Icons	Lucide React Icons	Vector UI icons for play, pause, volume, search, drawer, and upload modal
State Management	Zustand 4.5	Global store managing current track, playlist queue, volume, active engine
Backend Runtime	Node.js Serverless	Vercel Serverless Functions handling search, resolution, and audio proxies
Database ORM	Prisma ORM 5.x	Type-safe database client and migration manager for relational schemas

Relational Database	SQLite Database	File-based relational database storing Users, Tracks, Playlists, and Liked Tracks
Native Mobile OS	Android SDK 34 (Java 17)	Native Android Studio project container using WebView, Java, and Gradle 8.13

3. Universal Music Search & Streaming Engines

Jiya Music integrates four specialized music discovery and audio streaming data providers:

- **JioSaavn HQ API Engine:** Fetches direct high-quality 320kbps MP3 audio stream URLs for Indian, Bollywood, Regional, and International catalog tracks.
- **YouTube & Invidious Audio Engine:** Acts as a universal fallback streamer for any track, live performance, or cover video available on YouTube.
- **Apple iTunes Search API:** Retrieves high-resolution cover art (600x600), release year, album metadata, and official preview snippets.
- **Spotify API Engine:** Provides global charts, trending music lists, and artist metadata.
- **Stream Proxy Pipeline (/api/stream):** Bypasses cross-origin (CORS) restrictions and stream header blocks to deliver seamless continuous playback.

4. Relational Database Architecture (SQLite / Prisma)

Data Model	Key Fields	Description & Relationships
User	id, email, phone, name, avatarUrl, isPro	Represents registered app users; holds 1-to-many relationship with Playlists and Tracks
Track	id, title, artist, album, genre, duration, audioUrl, coverUrl, playCount	Stores custom user uploaded tracks and cached search tracks; linked to Playlists
Playlist	id, title, description, coverUrl, isPublic, userId	User-created music playlists containing custom track collections
PlaylistTrack	id, playlistId, trackId, position, addedAt	Junction table linking tracks to playlists with position ordering
LikedTrack	id, userId, trackId, createdAt	Stores favorite/liked songs for quick user library access

5. Native Android Mobile Architecture (/android-app)

To deliver a native mobile experience on Android devices without relying on web browser limitations, a standalone Android Studio project container was built inside /android-app:

Android Component	Java / System Class	Functionality & Purpose
Main Container Activity	MainActivity.java	Hosts WebKit WebView container with JavaScript enabled, DOM storage, and custom WebViewClient
Native Audio Engine	MediaPlayer & AudioAttributes	Native C++/Java audio player executing outside browser DOM to bypass WebView focus pause
Background Service	AudioService.java	Foreground Service with ongoing system notification to prevent Android OS process death
CPU WakeLock	PARTIAL_WAKE_LOCK	Maintains active CPU decoding state when phone screen is turned off or locked
JS-Native Bridge	@JavascriptInterface (AndroidNativePlayer)	Bridges web click events to native Java playNativeAudio(url, title, artist)
Native File Chooser	onShowFileChooser	Overridden WebChromeClient handler enabling native Android file picker for track uploads

6. Key Features & Engineering Highlights

- **Uninterrupted Screen-Off Playback:** Native Android Foreground Service and `PARTIAL_WAKE_LOCK` ensure music plays continuously when switching apps or locking the screen.
- **Lockscreen & Notification Media Controls:** Integrated `navigator.mediaSession` API showing song title, artist name, cover art, play/pause, and skip controls on Android lock screens.
- **Track Upload Studio:** Built-in modal allowing users to upload custom `.mp3`, `.wav`, and `.flac` files along with cover artwork directly to the server.
- **Instant Web Update Sync:** Any web app code or design update pushed to GitHub automatically syncs to all installed Android APK apps without requiring users to reinstall.
- **Dynamic Watchdog Fallback:** 1.5-second stalled playback watchdog automatically resolves alternative audio stream links if a stream fails or stalls at 0:00.

Report Generated: Jiya Music Engineering Team • Full Stack & Mobile Documentation